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CUET UG

Physics (322)

Ray Optics & Optical Instruments

Official Syllabus Unit 6 · Class 12 NCERT · CUET UG 2027 Cycle Notes

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Syllabus Alignment

Sub-topic	Official syllabus pointer (NTA Physics-322, Unit 6)	Skills the questions test
Reflection of light	Spherical mirrors, mirror formula, sign convention (Cartesian), magnification $m = -\frac{v}{u}$.	Sign discipline, image-position recall, magnification sign reading.
Refraction of light	Refraction through a glass slab, Snell's laws, total internal reflection and its critical angle, applications to optical fibres and prisms.	Applying $n_1 \sin\theta_1 = n_2 \sin\theta_2$, apparent-depth reasoning, TIR conditions.
Refraction at spherical surfaces & lenses	Thin lens formula, lensmaker's formula, power $P = \frac{1}{f}$ (dioptre), combination of thin lenses in contact, magnification.	Sign-flip awareness versus mirrors, dioptre arithmetic, medium-change effects.
Prisms	Refraction and dispersion through a prism, angle of minimum deviation $n = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$, thin-prism relation $\delta \approx (n-1)A$.	Index from δ_m, thin-prism shortcut, VIBGYOR ordering by n.
Optical instruments	Simple microscope (magnifying glass), compound microscope, astronomical telescope: magnifying powers and tube lengths at normal adjustment.	Formula recall for M and L, objective-versus-eyepiece role questions.

EXAM PATTERN

CUET UG pattern (2026 bulletin, the current 2027 baseline): each subject paper has 50 compulsory MCQs in 60 minutes, +5 for every correct answer and −1 for every incorrect answer. Unit 6 (Ray Optics) is tested through single-correct MCQs at NCERT level — every technique below is paired with its fastest exam-ready version.

Reflection at Spherical Mirrors

A spherical mirror is a small slice cut from a hollow glass sphere, polished on one side. When the **inner** (caved-in) surface does the reflecting, the mirror is **concave** and it brings parallel rays together — it **converges** light. When the **outer** (bulging) surface reflects, the mirror is **convex** and it spreads rays apart — it **diverges** light. Before any formula, get comfortable with the six location words that every mirror question is built on; each one is just a named point on the axis of the mirror.

- **Pole (P):** the exact centre of the mirror surface. All distances in problems are measured from here.

- **Centre of curvature (C):** the centre of the original sphere from which the mirror was cut. For a concave mirror it sits in front of the mirror; for a convex mirror, behind it.
- **Radius of curvature (R):** the distance P to C. A gently curved mirror has a large R; a strongly curved one has a small R.
- **Focus (F):** rays arriving parallel to the axis reflect through (concave) or appear to come from (convex) this point. It always sits exactly midway between P and C.
- **Focal length (f):** the distance P to F, so $f = \frac{R}{2}$ always. This one line converts "curvature" language into formula language.
- **Aperture:** the actual width of the mirror. For all school-level work the mirror is assumed thin (small aperture), which is what makes $f = \frac{R}{2}$ exact.

The mirror sits on a **principal axis** — the straight line through P and C. Distances are measured from P using the **Cartesian sign convention**: take the incoming light as travelling left to right, call distances measured **against** it negative and distances **behind** the mirror positive, and call heights above the axis positive. Fix this convention once and every mirror problem becomes careful bookkeeping instead of guesswork; a single sign slip is what flips a final answer from real to virtual.

DIAGRAM **Image Formation by a Concave Mirror — Object Beyond C**

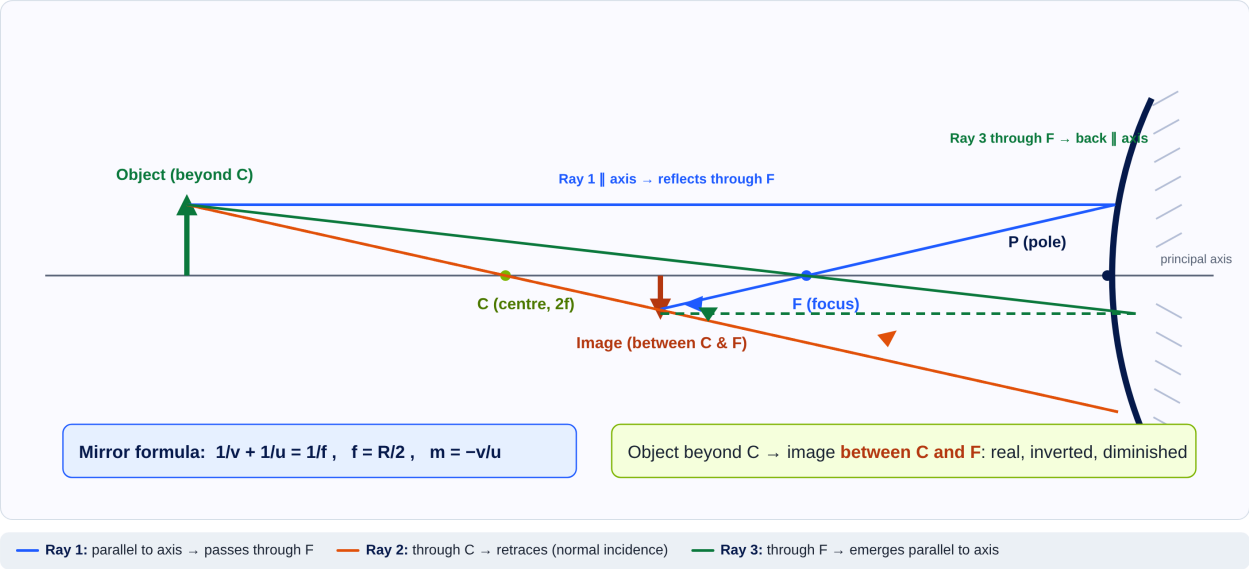


Figure 1 — The three standard rays used to construct any concave-mirror image.

Any image location can be found by drawing just two of these three rays, and the third is a free cross-check:

- **Parallel ray:** a ray parallel to the axis reflects through the focus F.
- **C-ray:** a ray aimed at the centre of curvature C hits the surface head-on (along the normal), so it reflects straight back along its own path.
- **Pole ray:** a ray striking the pole P reflects at the same angle on the other side of the axis, like a ball bouncing off a flat wall.

KEY FORMULA

$$\text{Mirror formula:} \quad \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\text{Radius relation:} \quad f = \frac{R}{2}$$

$$\text{Magnification:} \quad m = -\frac{v}{u} = \frac{h'}{h}$$

Read the signs as physics, not arithmetic. A **negative** image distance v means the image is on the object's side of the mirror — a **real** image that can be caught on a screen. A **positive** v means the image is behind the mirror — **virtual**, visible to the eye but not projectable. The magnification sign carries the orientation: negative m means inverted (and therefore real for a mirror), positive m means erect (virtual). For a concave mirror f is negative in this convention; a convex mirror always has f positive and therefore can only ever produce virtual, diminished, erect images — a one-line fact worth memorising.

Where the image forms: the concave-mirror map

Object position	Image position	Image nature and everyday use
At infinity	At the focus F	Point-sized, real, inverted — solar furnace collects this
Beyond C	Between F and C	Real, inverted, diminished — torch and headlight reflectors
At C	At C itself	Real, inverted, same size — classic "image at the centre" question
Between F and C	Beyond C	Real, inverted, enlarged — projector arrangement
At F	At infinity	Reflected rays leave parallel — searchlight geometry
Between P and F	Behind the mirror	Virtual, erect, enlarged — shaving and make-up mirror

Each row of this map is worth memorising: cover the right two columns, read the object position, and recall the image position and nature from memory. The convex mirror column is much shorter for a reason — no matter where the object is, a convex mirror always forms a virtual, erect, diminished image between P and F. That exact behaviour is why it is chosen for vehicle rear-view mirrors and shop security domes: a wider field of view fits into a small, always-upright image.

EVERYDAY EXAMPLE — THE SOUP SPOON

Turn a shiny spoon towards you with its **inner** (bowl) side: your reflection is upside-down and shrunken or blown-up depending on how far you hold it — that bowl is a concave mirror. Flip it to the **outer** back side: your reflection is small, upright and slightly distorted — a convex mirror. One object, both mirrors, and the full sign convention in your kitchen drawer.

WORKED EXAMPLE — MIRROR FORMULA

An object stands 30 cm in front of a concave mirror of focal length 20 cm. Locate the image and find its magnification.

Solution. Apply the convention: $u = -30$ cm, $f = -20$ cm.

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{-20} - \frac{1}{-30} = -\frac{1}{60} \quad \Longrightarrow \quad v = -60 \text{ cm}$$

The negative v places the image 60 cm in front of the mirror — real, and beyond C (which sits at $2f = 40$ cm). That matches the map: the object is between F and C, so the image lands beyond C, enlarged. Magnification: $m = -\frac{v}{u} = -\frac{-60}{-30} = -2$ — inverted and twice the object height. The signs did all the conceptual work.

WORKED EXAMPLE — CONVEX REAR-VIEW MIRROR

A convex mirror has focal length 15 cm. An object is 30 cm away. Where is the image, and what is the magnification?

Solution. $u = -30$ cm, $f = +15$ cm:

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{15} + \frac{1}{30} = \frac{1}{10} \quad \Longrightarrow \quad v = +10 \text{ cm}, \quad m = -\frac{v}{u} = +\frac{1}{3}$$

Positive v — the image is behind the mirror, virtual and erect; $m = +\frac{1}{3}$ — three times smaller. The driver sees a wide, upright, reduced view, which is exactly why the mirror carries the warning “objects are closer than they appear”.

Refraction: Why Light Bends at a Boundary

Drop a pencil into a glass of water and look from the side: the pencil appears bent at the waterline even though it is straight. The pencil did not bend — the light coming from the underwater part **changed direction** at the surface. That direction change is **refraction**, and its single cause is a **change in the speed of light** when it crosses from one transparent medium into another. Light travels at $c \approx 3 \times 10^8$ m/s in vacuum and slower inside matter: about 2.25×10^8 m/s in water and 2×10^8 m/s in glass. The medium’s **refractive index** n is simply the slowdown factor:

REFRACTIVE INDEX AND SNELL'S LAW

$$n = \frac{c}{v}$$

$$\text{Snell's law:} \quad n_1 \sin\theta_1 = n_2 \sin\theta_2$$

$$\text{Apparent depth:} \quad n = \frac{\text{real depth}}{\text{apparent depth}}$$

Three facts follow directly from the speed picture, and each is a common one-mark check:

- **Frequency never changes.** The source sets how many wave crests arrive per second; crossing a boundary cannot change that count.
- **Wavelength shrinks inside the denser medium** to $\frac{\lambda}{n}$, since $v = f\lambda$ and f is fixed while v drops.
- **Bending direction follows the speed change.** Entering a denser (slower) medium, the ray bends **towards** the normal; entering a rarer (faster) one, it bends **away**.
- **The path is reversible.** If a ray is reversed at the end of its journey, it retraces the same bends in reverse — this symmetry lets you swap n_1 and n_2 freely when checking an answer.
- **No boundary, no bend.** A ray striking the surface along the normal ($\theta_1 = 0°$) changes speed but does not change direction — bending needs an oblique angle.

Memorise a few anchor values of n and most ranking questions answer themselves:

Medium	Refractive index n	Speed of light inside (approx.)
Vacuum	1 (exactly)	3×10^8 m/s
Air	1.0003 (treated as 1)	essentially c
Water	1.33	2.25×10^8 m/s
Crown glass	1.5	2×10^8 m/s
Diamond	2.42	1.24×10^8 m/s — the slowest common medium

The larger the index, the slower light travels and the more a ray bends on entry. Diamond tops this list among everyday materials, which is why it both sparkles (large bending, strong trapping) and ranks highest in every "arrange by bending" question. Optical density is a statement about light speed, not about mass density — petrol floats on water but has a slightly higher refractive index than water, a classic trap pairing.

REAL-WORLD ANALOGY

A marching band walks from smooth pavement onto a sandy patch **at an angle**. The marchers who reach the sand first slow down while their row-mates on pavement keep pace, so the entire row pivots — the band's line of march kinks at the boundary. Each marcher is a wavefront point; the pavement-to-sand slowdown is the vacuum-to-glass slowdown, and the kinked path is exactly Snell's law.

Snell's law at work: pools, coins and glass slabs

The **apparent depth** relation explains why a swimming pool looks shallower than it really is: light from the floor bends away from the normal as it leaves the water, so the brain projects the rays along straight lines and places the floor higher than it is. Water with $n = \frac{4}{3}$ makes a 3 m deep pool appear only $\frac{3}{\frac{4}{3}} = 2.25$ m deep. The same geometry means a coin at the bottom of a bucket appears closer to the rim — and a spear-fisher must aim **below** the seen fish to catch it.

A parallel-sided glass slab refracts twice — once on entry, once on exit. The two bends exactly cancel for direction, so the emergent ray is **parallel** to the incident ray; but it is laterally shifted

by a few millimetres. Direction preserved, position shifted: that is why text seen through a thick glass pane looks in place but slightly displaced sideways when you move your eye. There is **no net deviation** in a slab, while a prism (next sections) deviates permanently — a contrast questions directly test.

WORKED EXAMPLE — SNELL'S LAW CALCULATION

Light passes from air ($n_1 = 1.0$) into glass ($n_2 = 1.5$) hitting the surface at 40° to the normal. Find the refraction angle and the speed of light inside the glass.

Solution. Snell's law: $\sin\theta_2 = \frac{1.0 \cdot \sin 40^\circ}{1.5} = \frac{0.643}{1.5} = 0.4285$, so

$$\theta_2 = \sin^{-1}(0.4285) \approx 25.4^\circ$$

The angle shrinks from 40° to 25.4° — bent **towards** the normal, as expected on entering a slower medium. Inside the glass the speed is $v = \frac{c}{n} = \frac{3\times10^8}{1.5} = 2 \times 10^8$ m/s.

Total Internal Reflection: When Light Cannot Escape

Stand underwater on a bright day and look up: you see the whole sky compressed into a bright circle overhead, and outside that circle the surface looks like a mirror showing the pool floor. That mirror is not a coating — it is **total internal reflection (TIR)**, and it happens because light travelling from a **denser** to a **rarer** medium bends **away** from the normal. Push the incidence angle far enough and the refracted ray lies flat along the surface at the **critical angle** θ_c; beyond it, refraction has nowhere to go and the light is reflected back **completely** — 100% of it, no partial reflections involved.

Two conditions must hold **simultaneously**, and both matter equally:

- **Right direction:** light must travel from the denser medium towards the rarer one (glass to air, water to air) — never the reverse.
- **Beyond the critical angle:** the incidence angle must exceed θ_c, where $\sin\theta_c = \frac{n_2}{n_1}$ (for a medium-to-air boundary, $\sin\theta_c = \frac{1}{n}$).

CRITICAL ANGLE

$$\sin\theta_c = \frac{n_2}{n_1}$$

$$\text{Medium to air:} \quad \sin\theta_c = \frac{1}{n}$$

$$\text{Glass-air (n = 1.5):} \quad \theta_c = \sin^{-1}\left(\frac{1}{1.5}\right) \approx 41.8^\circ$$

$$\text{Water-air:} \quad \theta_c \approx 48.75^\circ$$

WORKED EXAMPLE — FINDING THE CRITICAL ANGLE

A transparent liquid has refractive index 1.33. Find its critical angle with air, and state what happens to a ray hitting the surface at 50° from inside.

Solution. $\sin\theta_c = \frac{1}{1.33} = 0.752$, so $\theta_c = \sin^{-1}(0.752) \approx 48.75°$. The ray arrives at 50°, which is **beyond** θ_c, and it is travelling denser to rarer — both TIR conditions hold, so the ray is totally internally reflected back into the liquid. Had the angle been 45°, partial refraction with a bent refracted ray would occur instead.

Watch the pattern: one wrong option in such MCQs violates the direction condition, another violates the angle condition. Check both before answering.

TIR is not a curiosity — it moves the world's information. Every optical fibre works by keeping light trapped: light entering the thin glass core hits the core-cladding boundary at angles beyond the critical angle and bounces along in a zig-zag with essentially zero loss, which is how undersea cables carry internet traffic across oceans as pulses of light. An endoscope uses the same trick to pipe light into and images out of the human body. Diamonds owe their sparkle to TIR: with the unusually high $n = 2.42$, the critical angle is only $\sin^{-1}\left(\frac{1}{2.42}\right) \approx 24.4°$, so light entering a cut stone bounces many times inside before escaping through the top — engineered facets make the trapping as long as possible. Mirages work in reverse: hot road air (rarer, near the ground) refracts and finally totally reflects the downward ray from the sky back up into your eye, painting a false "wet" pool of sky on the road.

The fibre works only because its **core** has a slightly higher refractive index than its **cladding** — TIR needs denser-to-rarer, so the boundary must face the right way. This direction rule also powers a family of glass prisms: a $45° - 90° - 45°$ prism receives light on one face at exactly 45°, which exceeds the 41.8° critical angle of glass, and turns the beam by 90° with zero silvering and zero loss. Two such prisms in a binocular fold a long light path into a compact body **and** re-erect the inverted image; a single one in a periscope lets a soldier see over a wall. Prism "mirrors" outperform metal mirrors because no metal coating exists to scratch, tarnish or absorb — the reflection is built into geometry itself.

- **Higher n means smaller θ_c:** diamond ($n = 2.42$) traps light far more aggressively than glass ($n = 1.5$) or water ($n = 1.33$) — ranking questions on θ_c reduce to ranking n in reverse.
- **TIR is total:** ordinary mirror reflections leave a few percent of light behind at every bounce; TIR returns essentially everything, which is why fibre optics can survive thousands of bounces per kilometre.
- **Frequency and colour are unchanged by TIR** — only direction changes, so trapped light keeps its message intact.

EVERYDAY EXAMPLE — THE POND-SKATER'S WINDOW

From **under** water, look up at a shallow angle: the water surface, which behaved like a clear window when viewed straight up, suddenly behaves like a perfect silver mirror. A swimmer sees the whole world above compressed into a 97° cone ($2 \times 48.75°$), and beyond that cone — the pond bottom reflected. TIR flips a window into a mirror purely because of viewing angle, no coating required.

Thin Lenses and the Lensmaker's Formula

A **lens** is a transparent piece of glass or plastic bounded by two curved surfaces (or one curved and one flat). A **convex** (converging) lens is fat in the middle: it bends parallel rays inward to a real focus, the way a magnifying glass concentrates sunlight into a bright, hot point. A **concave** (diverging) lens is thin in the middle and spreads rays outward so they appear to come from a point **behind** the lens — a virtual focus. Because a lens refracts at **two** surfaces, its sign story differs from a mirror's in exactly one place, and that one place decides marks:

Quantity	Spherical mirror	Thin lens
Master formula	$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$ ("plus")	$\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$ ("minus")
Real image distance	Negative ($v<0$, same side as object)	Positive ($v>0$, opposite side of object)
Converging element	Concave mirror, $f<0$	Convex lens, $f>0$
Diverging element	Convex mirror, $f>0$	Concave lens, $f<0$

A reliable way to hold this table: for mirrors, "real is on the object side" (so v negative); for lenses, real images form on the **transmission** side (so v positive). The formula difference — plus for mirrors, minus for lenses — is not arbitrary; it is the algebraic shadow of exactly this geometric difference.

POWER OF A LENS

$$P=\frac{1}{f}\ \text{(f in metres)}$$

Dioptre: $1\ \text{D}=1\text{m}^{-1}$

Lenses in contact: $P=P_1+P_2+\ldots$

Equivalent focal length: $\frac{1}{F}=\frac{1}{f_1}+\frac{1}{f_2}+\ldots$

Power is the practical language of optometry. An optician's prescription reading -1.5 D means a diverging (concave) lens of focal length $-\frac{1}{1.5}\approx-67$ cm for a myopic (short-sighted) eye; $+2.0$ D means a converging lens of focal length $+50$ cm for a hypermetropic (long-sighted) reader. Powers of stacked thin lenses simply **add**, so an optometrist can correct a combination of defects by summing dioptres — the arithmetic a phoropter dial performs while the patient reads letters.

LENSMAKER'S FORMULA

$$\frac{1}{f}=(n-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$$

$n=$ lens material index, $\quad R_1, R_2=$ radii of the two surfaces

The lensmaker's formula tells you f is decided by two independent choices: the **material** (bigger $n-1$ gives a shorter, stronger focus — flint glass bends more than crown glass) and the **shape** (smaller radii, i.e. more curved surfaces, shorten f). It also predicts what happens when the **surrounding medium** changes. An air bubble inside water acts as a **diverging** lens even though its glass-free shape looks converging: here the "lens material" is air ($n = 1.0$) and the surroundings are water ($n = 1.33$), so the effective $(n-1)$ is negative. In the same spirit, a converging glass lens immersed in carbon disulphide ($n = 1.63 > 1.5$) flips character and starts diverging. Any MCQ of the form "a lens behaves differently inside a liquid" is this one paragraph.

WORKED EXAMPLE — TWO LENSES IN CONTACT

Two thin lenses of focal lengths $+15$ cm and -10 cm are placed in contact. Find the focal length and power of the combination, and state its behaviour.

Solution. $\frac{1}{F} = \frac{1}{15} + \frac{1}{-10} = \frac{2}{30} - \frac{3}{30} = -\frac{1}{30}$, so $F = -30$ cm and $P = \frac{1}{-0.30\ \text{m}} = -3.33$ D. The pair behaves as a single diverging lens of focal length 30 cm. The same arithmetic nets a contact-lens prescription to a target power: choose components whose dioptres sum to the required value.

WORKED EXAMPLE — IMAGE BY A CONVERGING LENS

An object is placed 36 cm from a convex lens of focal length 12 cm. Find the image position and magnification.

Solution. $u = -36$ cm, $f = +12$ cm:

$$\frac{1}{v} = \frac{1}{f} + \frac{1}{u} = \frac{1}{12} - \frac{1}{36} = \frac{3-1}{36} = \frac{1}{18} \quad \Longrightarrow \quad v = +18 \text{ cm}$$

Positive v — a real image on the far side, 18 cm from the lens. The signed magnification is $m = \frac{v}{u} = \frac{18}{-36} = -\frac{1}{2}$: inverted and half-size (for a single thin lens, every real image is inverted — the negative sign is not optional bookkeeping, it is the physics). When an object sits beyond $2f$, a converging lens always delivers a real, inverted, diminished image between f and $2f$ — the operating regime of a camera.

The camera connection: your phone camera's "zoom" is partly this formula at work — changing the effective f moves where the sharpest real image lands on the sensor.

EVERYDAY EXAMPLE — THE READING STONE

A magnifying glass starts a fire because it maps the huge, nearly parallel disc of sunlight onto one tiny bright spot — all the energy passing through the whole lens lands in a millimetre-sized real image of the sun. That spot **is** the focal point doing its job: a converging lens forming a real image at its focus. The same physics, used gently, magnifies a page instead.

Prisms, Dispersion and Minimum Deviation

A prism is a wedge of transparent material with two flat refracting faces meeting at the **angle of the prism** A. A ray entering it is bent twice — towards the normal on entry, away from it on exit

— and the two bends do **not** cancel because the faces are not parallel. The total turn between the incoming and outgoing ray is the **angle of deviation** $\delta = i + e - A$, where i and e are the external incidence and emergence angles. As you slowly rotate the prism (equivalently, vary i), δ first decreases, reaches a single minimum value δ_m, then rises again. At that special setting the ray inside the prism travels **parallel to the base** and the entry and exit angles match ($i = e$) — a beautiful symmetry forced by the geometry.

PRISM RELATIONS

$$\delta = i + e - A \qquad A = r_1 + r_2$$

$$\text{At minimum deviation:} \quad n = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

$$\text{Thin prism:} \quad \delta \approx (n-1)A$$

The minimum-deviation formula is the workhorse: measure A and δ_m in a lab and you have measured the refractive index without knowing any radii. The thin-prism shortcut handles the small-angle limit used in spectrometers: deviation grows linearly with both n and prism angle, no sines required. Keep the validity border in mind — $\delta \approx (n-1)A$ is for small A only; at $A = 60°$ you must use the full sine form.

Dispersion: why prisms paint rainbows

Send white light through a prism and a **spectrum** fans out — violet through red. This splitting is **dispersion**, and its cause is that n is not one number but a **function of wavelength**: violet light slows more in glass than red light does, so $n_{\text{violet}} > n_{\text{red}}$, so violet bends more. The two ends of the visible spectrum part ways inside the prism and leave it spread by the **angular dispersion** $(n_{\text{violet}} - n_{\text{red}})A$ for a thin prism. Order from top (least deviated) to bottom (most deviated) of a spectrum on a screen: **red, orange, yellow, green, blue, indigo, violet** — red bends least, violet most.

Setup	Direction of ray	Dispersion and deviation
Parallel glass slab	Emerges parallel to incident ray	Splits colours inside but they recombine on exit: dispersion **without** net deviation
Thin prism	Net bend of $(n-1)A$	Permanent deviation **and** permanent angular spread of colours
Prism at minimum deviation	Symmetric passage, ray parallel to base	Cleanest single-deviation setting; the n-measurement condition

This table answers a classic conceptual pair: "does a glass slab disperse?" (yes, but the colours recombine — no lasting spectrum) and "why does a prism not?" (non-parallel faces lock the spread in). A **rainbow** is dispersion plus TIR on a grand scale: sunlight enters a raindrop, refracts, reflects

once off the back surface, refracts again on exit, and each wavelength exits at its own angle near 42° (red) to 40° (violet) from the antisolar point — millions of drops each painting one pixel of the arc you see.

If you plot measured deviation δ against incidence angle i for a fixed prism, the curve falls steeply, flattens near δ_m, and rises again asymmetrically — the flat minimum is why lab measurements taken near δ_m are forgiving of small aim errors. Three factors move δ_m up or down, and each is a one-line question: a **larger prism angle** A increases deviation; a **higher refractive index** increases deviation (flint glass, $n \approx 1.62$, deviates more than crown glass, $n \approx 1.5$, for the same geometry); and a **shorter wavelength** increases deviation because n itself is larger for violet than for red. Dark-line spectra from atoms, measured with precisely this apparatus, are how astronomers read the composition of stars — the prism is the oldest and still one of the most productive scientific instruments in existence.

WORKED EXAMPLE — INDEX FROM MINIMUM DEVIATION

A equilateral prism ($A = 60°$) shows minimum deviation $\delta_m = 30°$. Find its refractive index.

Solution. Substitute directly:

$$n = \frac{\sin\left(\frac{60° + 30°}{2}\right)}{\sin(30°)} = \frac{\sin 45°}{\sin 30°} = \frac{\frac{1}{\sqrt{2}}}{\frac{1}{2}} = \sqrt{2} \approx 1.414$$

Numbers are chosen so the half-angles land on standard sines — if your δ_m produces an unfamiliar half-angle, re-check your arithmetic before panicking.

WORKED EXAMPLE — THIN PRISM

A thin prism of angle 5° is made of glass with $n = 1.5$. Find the deviation, and the deviation for violet light with $n_{\text{violet}} = 1.52$.

Solution. $\delta \approx (n-1)A = 0.5 \times 5° = 2.5°$. For violet: $\delta_v \approx (1.52 - 1) \times 5° = 2.6°$. The angular spread between violet and mean is 0.1° — small, but multiplied across a spectrometer's scale it is easily measured, and across a rain-shower sky it becomes a rainbow.

Optical Instruments: Microscopes and Telescopes

Every optical instrument is a controlled arrangement of two jobs: one element **gathers and forms** an image, the second **magnifies** it for the eye. Keep the roles separate and the formulas organise themselves. The **magnifying power** M of any instrument is the ratio of the angle subtended by the final image at the eye to the angle the object would subtend viewed nakedly at the **least distance of distinct vision**, $D = 25$ cm — the closest distance a normal eye focuses comfortably.

- **Simple microscope (magnifying glass):** a single convex lens held so the object sits inside its focal length. The image is virtual, erect and enlarged at or beyond 25 cm. Magnifying power $M = 1 + \frac{D}{f}$ with the image at the near point, and simply $M = \frac{D}{f}$ with the final image relaxed at infinity. A watchmaker's loupe of $f = 5$ cm gives $M = 1 + \frac{25}{5} = 6$.
- **Compound microscope:** two short-focus convex lenses in a tube. The **objective** (f_o, very small) forms a real, inverted, magnified intermediate image just inside the eyepiece's

focus; the **eyepiece** (f_e, small) magnifies that intermediate image again like a magnifying glass. Magnifying powers **multiply**: $M = \left(\frac{L}{f_o}\right)\left(\frac{D}{f_e}\right)$ at normal adjustment, where L is the tube length (distance between the lenses' back focal points). A biology lab microscope achieving $M = 500$ is really two modest magnifications, roughly 25×20, multiplied.

- **Astronomical telescope:** the roles flip by scale. The **objective** has a **long** focal length and gathers starlight into a real image at its focus; the eyepiece (short focus) magnifies that image. At **normal adjustment** the final image sits at infinity and the tube length equals $L = f_o + f_e$, while $M = \frac{f_o}{f_e}$ in magnitude (negative sign records the inversion). A telescope with $f_o = 100$ cm and $f_e = 5$ cm gives $M = 20$ and a 105 cm tube.

MAGNIFYING POWERS (STANDARD SETTINGS)

$$\text{Simple magnifier:} \quad M = 1 + \frac{D}{f}$$

$$\text{Compound microscope:} \quad M = \left(\frac{L}{f_o}\right)\left(\frac{D}{f_e}\right)$$

$$\text{Telescope:} \quad M = -\frac{f_o}{f_e}, \quad L = f_o + f_e$$

Feature	Compound microscope	Astronomical telescope
Objective focal length	Very small (f_o a few mm)	Very large (f_o tens of cm to metres)
Object	Tiny, placed just beyond f_o	Effectively at infinity
Intermediate image	Real, inverted, enlarged	Real, inverted, diminished
Final image	Virtual, enlarged, at D or ∞	Virtual, at ∞ (normal adjustment)
Tube length	about L (small)	$L = f_o + f_e$
Everyday example	School biology lab microscope	Observatory refractor, binoculars (with prisms to re-erect the image)

The microscope-versus-telescope contrast compresses to one line: the microscope's objective has a **tiny** focal length because its object is centimetres away; the telescope's objective has a **huge** focal length because its object is light-years away. Both eyepieces behave identically — as magnifying glasses examining the intermediate image.

Magnification is only half of an instrument's story; **light gathering** is the other half. The amount of light an objective collects grows with its **area**, so a 10 cm aperture telescope gathers roughly $\left(\frac{10}{0.5}\right)^2 = 400$ times more light than the pupil of your eye — that, not magnification alone, is why faint galaxies become visible. This also explains the feature everyone notices first about binoculars: the specification "8×40" means magnification 8 with a 40 mm objective, and the big front lens matters as much as the zoom. Modern prism binoculars fit two telescopes into a hand by using

the 45° TIR prisms of the previous section to fold the optical path twice; the same reflections flip the inverted image upright, which is why binoculars show a correctly oriented world while a plain astronomical telescope shows an upside-down one.

- **Why the final image can be at infinity:** at normal adjustment the intermediate image forms exactly at the eyepiece's focus, so the rays leaving the eyepiece are parallel — the relaxed eye views them with zero strain. The "image at near point" setting (M larger by the $1 + \frac{D}{f}$ factor) trades comfort for a little extra magnification.
- **Why the objective comes first optically:** it decides both resolution and brightness; the eyepiece only enlarges what the objective captured, which is why swapping eyepieces changes zoom but not detail.

WORKED EXAMPLE — COMPOUND MICROSCOPE

A compound microscope has $f_o = 1$ cm, $f_e = 5$ cm and tube length $L = 20$ cm. Find the magnifying power at normal adjustment.

Solution. $M = \left(\frac{L}{f_o}\right)\left(\frac{D}{f_e}\right) = \left(\frac{20}{1}\right) \times \left(\frac{25}{5}\right) = 20 \times 5 = 100$. The objective magnifies 20 times, the eyepiece 5 times, and the product is the answer. If asked instead for the image at the near point, the eyepiece factor becomes $1 + \frac{D}{f_e} = 6$ and $M = 120$ — always confirm **where** the final image is meant to be before computing.

WORKED EXAMPLE — ASTRONOMICAL TELESCOPE

A telescope has an objective of focal length 100 cm and an eyepiece of focal length 5 cm. Find the magnifying power and tube length at normal adjustment, and state what change increases M.

Solution. $M = -\frac{f_o}{f_e} = -\frac{100}{5} = -20$: magnitude 20, inverted final image. $L = f_o + f_e = 105$ cm. From $M = \frac{f_o}{f_e}$, magnification rises with a **longer** objective or a **shorter** eyepiece — the reasoning behind the long barrels of refracting telescopes.

EVERYDAY EXAMPLE — PASSING THE BATON

Think of both compound instruments as a two-stage relay. In a microscope, stage one (objective) photographs a tiny object into a bigger intermediate picture; stage two (eyepiece) holds a magnifying glass over that photograph. In a telescope, stage one compresses a huge, distant scene onto a small but **sharp** intermediate picture, and stage two blows it up for your eye. Neither stage is magical alone — the multiplication of the two is the instrument.

Practice Questions — CUET UG Pattern (10 MCQs)

Attempt these under exam timing: about 60 seconds per question. Fully worked solutions follow in the next section, so commit to an answer before reading on. Every question maps to one of the six sub-topics covered above.

Q1. An object is placed at the centre of curvature of a concave mirror. The image formed is:

(A) at F
(B) at C, same size, inverted
(C) beyond C
(D) at infinity

Q2. A convex mirror of focal length f forms an image of an object at distance v from the mirror. Which statement is always true?

(A) $|v| < f$
(B) $|v| = f$
(C) $|v| > f$
(D) $|v|$ can exceed $2f$

Q3. Light travels from water ($n = 1.33$) into glass ($n = 1.5$). Which statement is correct?

(A) TIR is possible at this boundary
(B) It bends away from the normal
(C) It bends towards the normal and speeds up
(D) It bends towards the normal and slows down

Q4. The critical angle for a medium–air interface is 45°. The refractive index of the medium is:

(A) 1.5
(B) 1.33
(C) $\sqrt{2}$
(D) $\frac{1}{\sqrt{2}}$

Q5. A concave lens of focal length 20 cm forms a virtual image at 15 cm from the lens. The object distance is:

(A) 60 cm
(B) 30 cm
(C) 12 cm
(D) 7.5 cm

Q6. Two thin lenses of powers +4 D and −2 D are placed in contact. The focal length of the combination is:

(A) +50 cm
(B) −50 cm
(C) +25 cm
(D) −25 cm

Q7. A prism of angle 60° produces minimum deviation of 30°. Its refractive index is closest to:

(A) 1.41
(B) 1.5
(C) 1.33
(D) 1.62

Q8. Which colour of white light deviates the **most** while passing through a glass prism?

(A) Red
(B) Green
(C) Violet
(D) All equally

Q9. In an astronomical telescope in normal adjustment, $f_o = 100$ cm and $f_e = 5$ cm. The magnifying power and tube length are respectively:

(A) 20 and 105 cm
(B) 20 and 95 cm
(C) 500 and 105 cm
(D) 95 and 20 cm

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Q10. A thin prism of angle $5°$ made of glass with $n = 1.5$ deviates light by approximately:

(A) $2.5°$ **(B)** $5°$
(C) $7.5°$ **(D)** $0.5°$

Detailed Solutions

Sol. Q1. The object is at C, so $u = -2f = -R$. The mirror formula gives $\frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{-f} + \frac{1}{2f} = -\frac{1}{2f}$, hence $v = -2f$: the image forms at C itself, real and inverted, same size. Magnification $m = -\frac{v}{u} = -\frac{2f}{2f} = -1$.

Answer: (B)

Sol. Q2. A convex mirror always forms a virtual, erect, diminished image between P and F, no matter where the object is. Since F lies f behind the mirror, the image distance magnitude is always less than f.

Answer: (A)

Sol. Q3. Water to glass is rarer to denser: the ray bends **towards** the normal and slows down because $v = \frac{c}{n}$ decreases. TIR needs the denser-to-rarer direction, so it is impossible at this boundary.

Answer: (D)

Sol. Q4. $n = \frac{1}{\sin}\theta_c = \frac{1}{\sin}45° = \frac{1}{\frac{1}{\sqrt{2}}} = \sqrt{2} \approx 1.414$. Options A and B are the familiar glass and water values that do not match $45°$.

Answer: (C)

Sol. Q5. Concave lens: $f = -20$ cm; a virtual image sits behind the lens, so $v = -15$ cm. Then $\frac{1}{u} = \frac{1}{v} - \frac{1}{f} = -\frac{1}{15} + \frac{1}{20} = \frac{-4+3}{60} = -\frac{1}{60}$, giving $u = -60$ cm: the object is 60 cm from the lens.

Answer: (A)

Sol. Q6. Powers add: $P = 4 - 2 = +2$ D. Convert to metres before inverting: $F = \frac{1}{P} = 0.5$ m $= +50$ cm. The classic slip is dividing 100 by P without converting dioptres to metres.

Answer: (A)

Sol. Q7. At minimum deviation, $n = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)} = \sin 45\frac{°}{\sin}30° = \frac{\frac{1}{\sqrt{2}}}{\frac{1}{2}} = \sqrt{2} \approx 1.41$.

Answer: (A)

Sol. Q8. Violet has the highest refractive index in glass, so it bends the most and lands at the bottom of the spectrum. The full order is VIBGYOR from most to least deviated: red deviates least.

Answer: (C)

Sol. Q9. Magnitude $M = \frac{f_o}{f_e} = \frac{100}{5} = 20$ (the image is inverted; many papers drop the sign), and $L = f_o + f_e = 105$ cm. Check both numbers — one distractor pair always gets L wrong by subtracting.

Answer: (A)

Sol. Q10. Thin-prism relation: $\delta \approx (n-1)A = 0.5 \times 5^\circ = 2.5^\circ$. The shortcut is valid only for small prism angles; for $A = 60^\circ$ use the full minimum-deviation sine formula.

Answer: (A)

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